

VPEM Graph

Vulnerability Prioritization and Exposure Management

Cybersecurity Use Cases: Evolving from reactive patching to risk-based strategy

Based on: [pedroleitao-neo4j/cyber-vpem](https://github.com/pedroleitao-neo4j/cyber-vpem)

The Problem: Alert Fatigue

Modern organizations are overwhelmed by thousands of software flaws.

- **The spreadsheet trap:** Treating a critical flaw on an isolated server the same as one on an internet-facing gateway.
- **Reactive vs Proactive:** Wasting resources on theoretical risks instead of real-world threats.
- **Lack of Context:** Severity scores (CVSS) don't account for your specific environment.

What is VPEM?

It shifts the focus from "What is broken?" to "What could actually hurt us right now?"

- **Exposure:** Is the system public-facing or internal?
- **Threat Intel:** Is there an active exploit in the wild (CISA KEV)?
- **Criticality:** Is this a test server or a production "Crown Jewel" database?

Core Project Architecture

The project is structured into three execution layers:

1. **Data Ingestion**: Merging Asset, Vulnerability (NVD), and Threat (CISA KEV) data.
2. **Specific Use Cases**: Reachability, Impact, and Contextual Risk scoring.
3. **Other Analysis**: Advanced path traversal and security insights.

Data Integration Layers

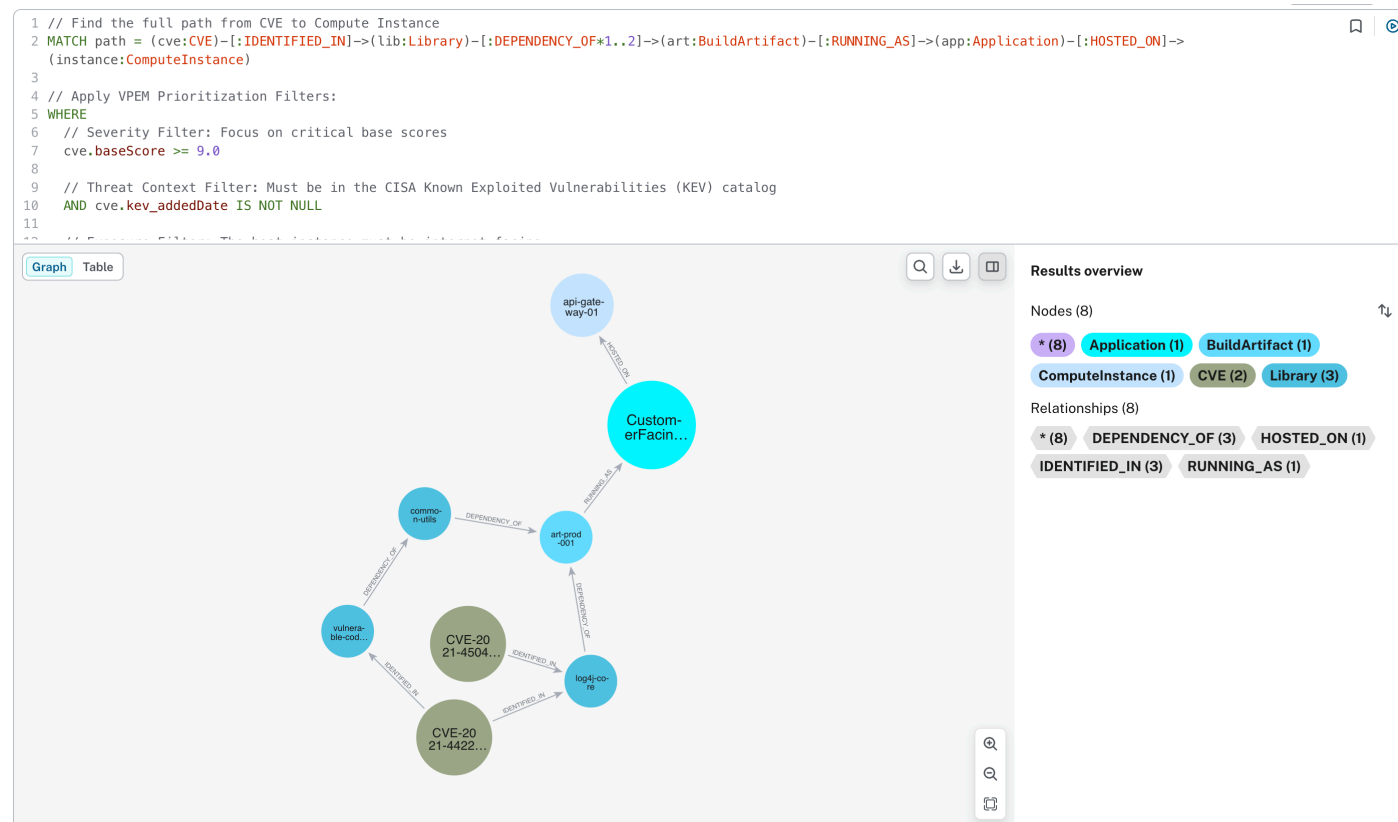
To build a holistic view, three distinct data domains are integrated:

Layer	Sources	Key Entities
Organizational Context	Cloud APIs, IAM, SBOMs	Apps, Compute, S3, IAM Policies
Vulnerability Intel	NVD	CVEs, CVSS Scores, Attack Vectors
Threat Intelligence	CISA KEV	Active Exploit Status, Deadlines

Visualizing Reachability

Vulnerability to Compute Instance Reachability

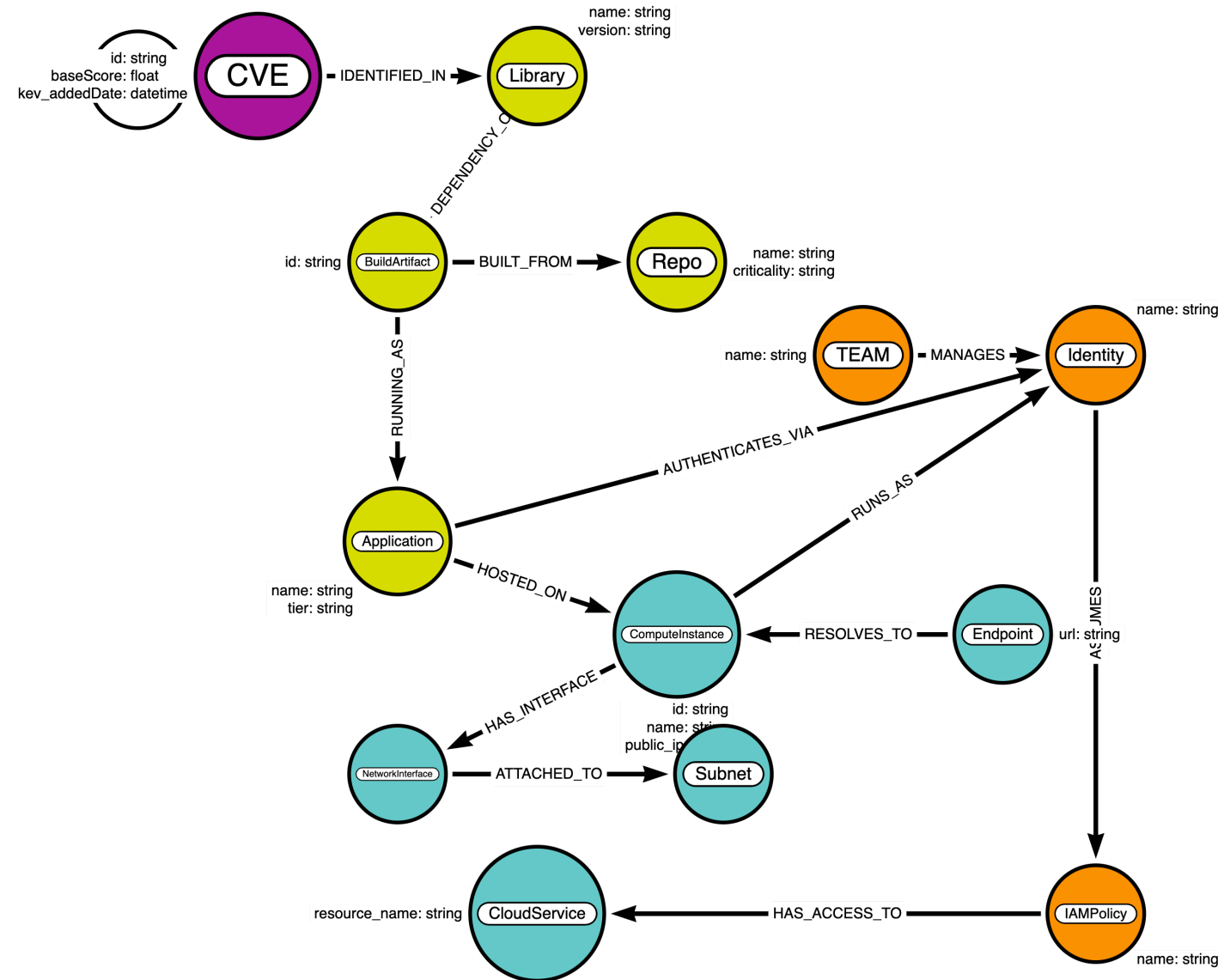
- **Attack Path Analysis:** Identifying the direct path from a CVE to an internet-facing host.
- **Blast Radius:** Mapping what a compromised identity can access.



The Resulting Schema

The graph connects code, infrastructure, and identity in real-time.

The extended schema captures deployment, usage, and vulnerability identification.



The Graph Advantage

Why use Neo4j for Vulnerability Management?

- **Real-world Reachability:** Is the vulnerable library actually reachable from an `Endpoint` ?
- **Impact Analysis:** If this server is compromised, which "Crown Jewel" assets (PII databases) are at risk via its Identity?
- **Efficiency:** Find the "Chokepoint"—the one library update that fixes the most reachable risk.

Outward System Integration

The VPEM graph acts as a **central nervous system** for security data:

- **SecOps (SOAR):** Contextual alert enrichment and automated containment.
- **DevOps (CI/CD):** "Policy as Code" guardrails to fail builds that create lateral movement paths.
- **GRC & Compliance:** Strategic risk metrics and CISA KEV remediation tracking.

Questions?

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